HDOC Day-43 Activity

Menu-Driven C Program - Longest Word and String Rotation

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Aim

To write a menu-driven C program using separate user-defined functions to find the longest word in a sentence and to check whether one string is a rotation of another.

Operations

1. Find the longest word in a sentence.
2. Check whether one string is a rotation of another.
3. Exit.

Concept Used

This program uses strings, arrays, functions, loops, switch-case, **strlen()**, **strcpy()**, **strcat()**, and **strstr()**. The header files used are **stdio.h** and **string.h**.

Algorithm

Longest word:

1. Read a complete sentence using fgets().
2. Scan the sentence character by character.
3. Count the length of each word.
4. Whenever a word ends, compare its length with the current maximum.
5. Store and display the longest word.

String rotation:

1. Read two strings.
2. If their lengths are different, they cannot be rotations.
3. Concatenate the first string with itself.
4. Search for the second string inside the doubled string using strstr().
5. If found, the second string is a rotation of the first string.

C Program

```
#include <stdio.h>
#include <string.h>

void findLongestWord(char sentence[])
{
    int i, start = 0, len = 0;
    int maxStart = 0, maxLen = 0;

    for (i = 0; ; i++)
    {
        if (sentence[i] != ' ' &&
            sentence[i] != '\n' &&
            sentence[i] != '\0')
        {
```

```
            if (len == 0)
                start = i;
            len++;
        }
        else
        {
            if (len > maxLen)
            {
                maxLen = len;
                maxStart = start;
            }

            len = 0;

            if (sentence[i] == '\0')
                break;
        }
    }

    if (maxLen == 0)
    {
        printf("No word found.\n");
        return;
    }

    printf("Longest word: ");
    for (i = maxStart; i < maxStart + maxLen; i++)
        printf("%c", sentence[i]);

    printf("\nLength: %d\n", maxLen);
}

int isRotation(char str1[], char str2[])
{
    char temp[400];

    if (strlen(str1) != strlen(str2))
        return 0;

    strcpy(temp, str1);
    strcat(temp, str1);

    if (strstr(temp, str2) != NULL)
        return 1;

    return 0;
}

int main()
{
    int choice;
    char sentence[200];
    char str1[100], str2[100];

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Find longest word in a sentence\n");
        printf("2. Check string rotation\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(sentence, sizeof(sentence), stdin);
                findLongestWord(sentence);
                break;

            case 2:
                printf("Enter first string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                printf("Enter second string: ");
```

```
                fgets(str2, sizeof(str2), stdin);
                str2[strcspn(str2, "\n")] = '\0';

                if (isRotation(str1, str2))
                    printf("The strings are rotations of each other.\n");
                else
                    printf("The strings are not rotations of each other.\n");
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Sample Output

```
----- MENU -----
1. Find longest word in a sentence
2. Check string rotation
3. Exit
Enter your choice: 1
Enter a sentence: C programming is interesting
Longest word: programming
Length: 11

Enter your choice: 2
Enter first string: ABCD
Enter second string: CDAB
The strings are rotations of each other.

Enter your choice: 2
Enter first string: ABCD
Enter second string: ACBD
The strings are not rotations of each other.

Enter your choice: 3
Exiting program...
```

Function Explanation

findLongestWord(): Scans the sentence, measures each word, keeps the starting position and length of the longest word, and finally prints that word.

isRotation(): First checks that both strings have equal lengths. It then creates **str1 + str1**. If str2 occurs inside this doubled string, str2 is a rotation of str1.

Easy Trick to Remember Rotation

Rotation formula: str1 + str1, then search str2.
Example: ABCD + ABCD = ABCDABCD. Since CDAB is present inside ABCDABCD, CDAB is a rotation of ABCD.

Result

Thus, the menu-driven C program was successfully implemented using separate functions to find the longest word in a sentence and check whether two strings are rotations of each other.

Quick Viva Questions

1. What is a string rotation? A string formed by moving characters from one end of a string to the other without changing their order.
2. Why must the lengths be equal? Rotation does not add or remove characters.
3. What does strstr() do? It searches for one string inside another string.
4. What does strlen() return? The number of characters in a string, excluding the null character.
5. Why is fgets() used for the sentence? It reads a complete line including spaces.
6. Give one rotation example. CDAB is a rotation of ABCD.